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**Wang et al.**

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(54) **ENHANCING A LISTENING EXPERIENCE BY ADJUSTING PHYSICAL ATTRIBUTES OF AN AUDIO PLAYBACK SYSTEM BASED ON DETECTED ENVIRONMENTAL ATTRIBUTES OF THE SYSTEM'S ENVIRONMENT**

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This patent is subject to a terminal disclaimer.

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**Related U.S. Application Data**

(63) Continuation of application No. 16/932,549, filed on Jul. 17, 2020, now Pat. No. 11,425,490, which is a continuation of application No. 16/355,128, filed on Mar. 15, 2019, now Pat. No. 10,735,854, which is a continuation of application No. 15/657,844, filed on Jul. 24, 2017, now Pat. No. 10,237,644.

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**H04R 3/04** (2006.01)  
**H04R 29/00** (2006.01)

(52) **U.S. Cl.**

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(58) **Field of Classification Search**

CPC .... **H04R 1/345**; **H04R 1/1091**; **H04R 1/2803**; **H04R 3/04**; **H04R 29/00**; **H04R 2460/07**  
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See application file for complete search history.

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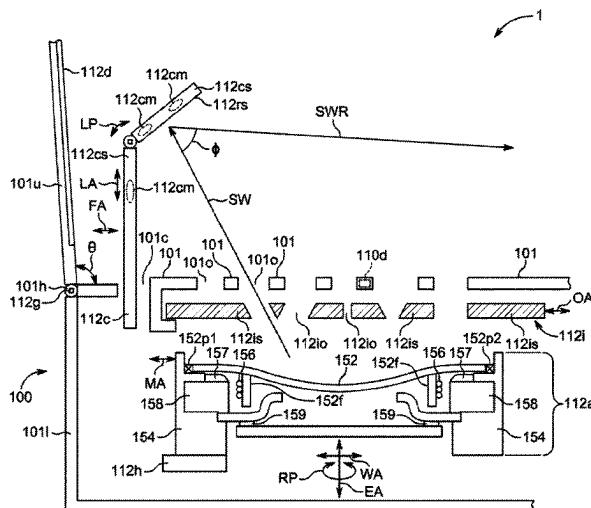
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**ABSTRACT**

Systems, methods, and computer-readable media are provided for enhancing a user's listening experience by adjusting physical attributes of an audio playback system based on detected environmental attributes of the system's environment.

**19 Claims, 7 Drawing Sheets**



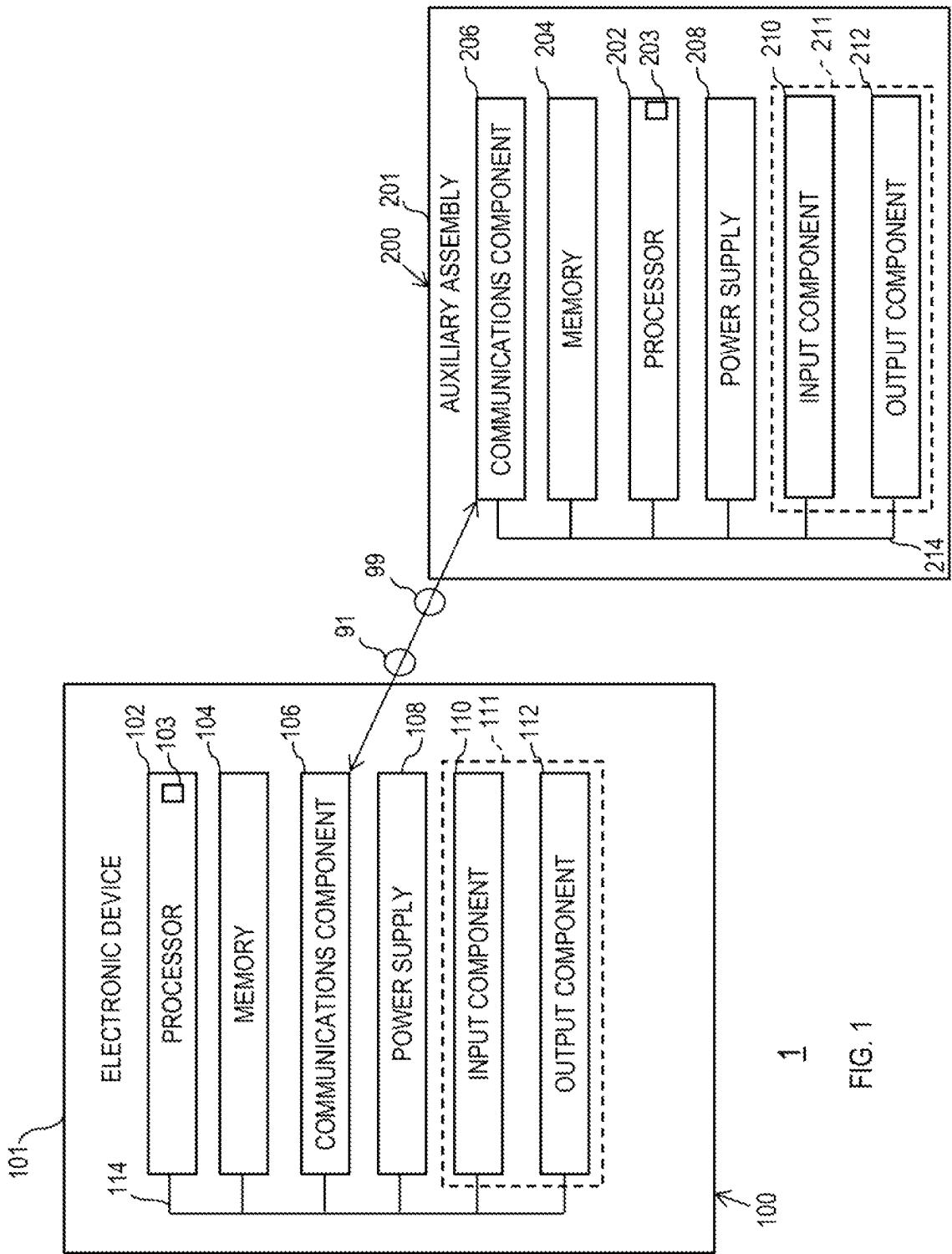
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FIG. 1



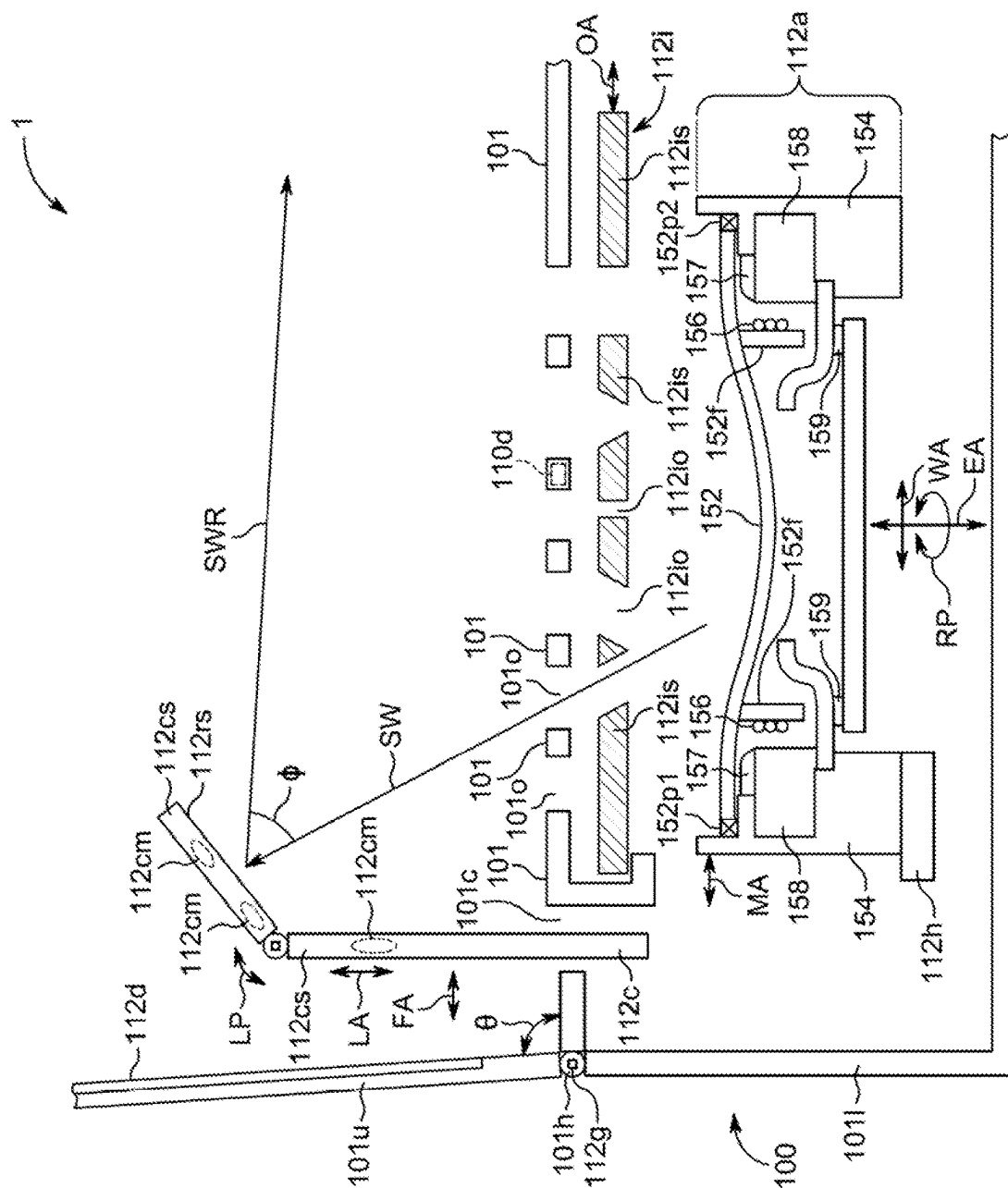
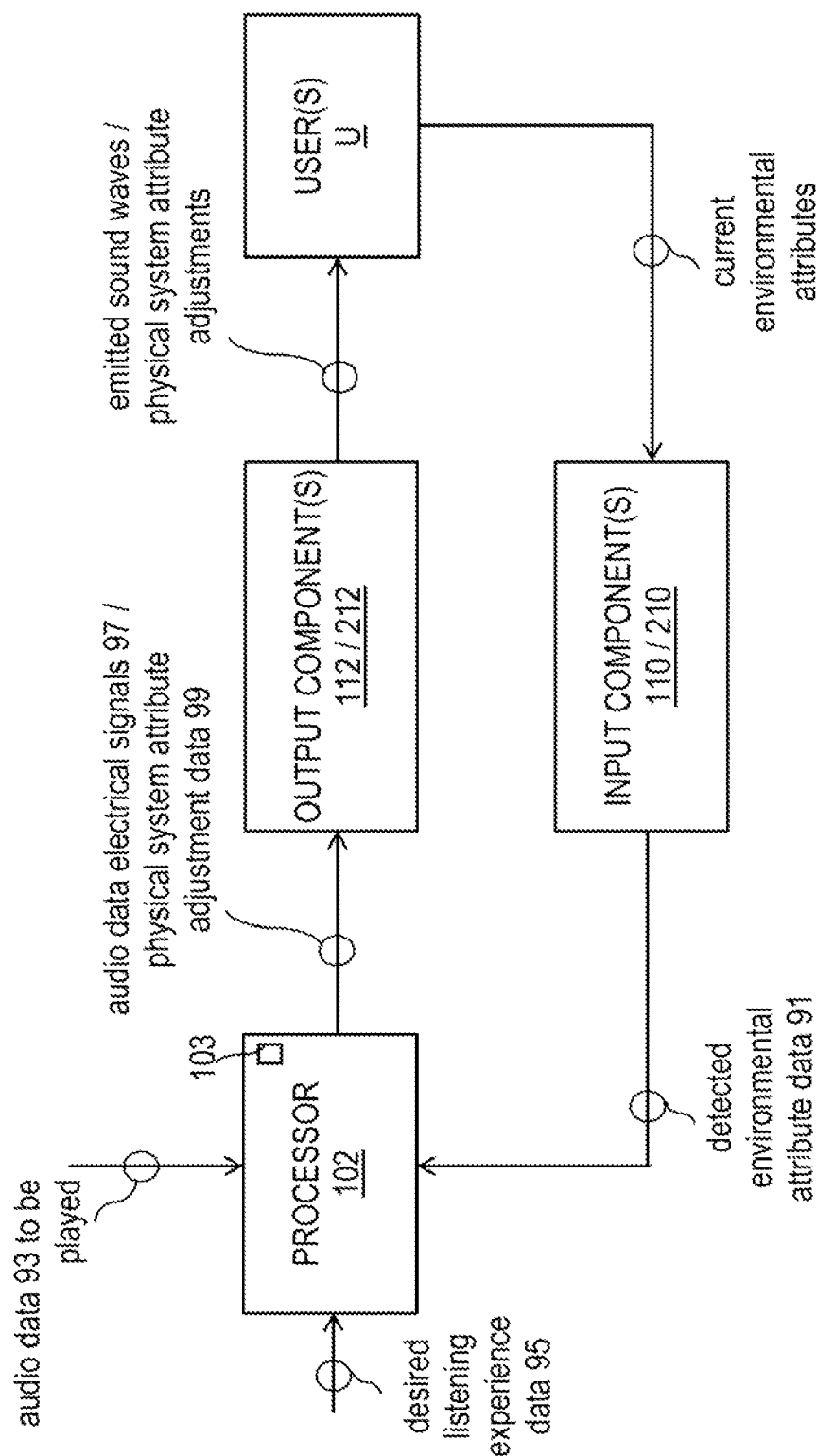
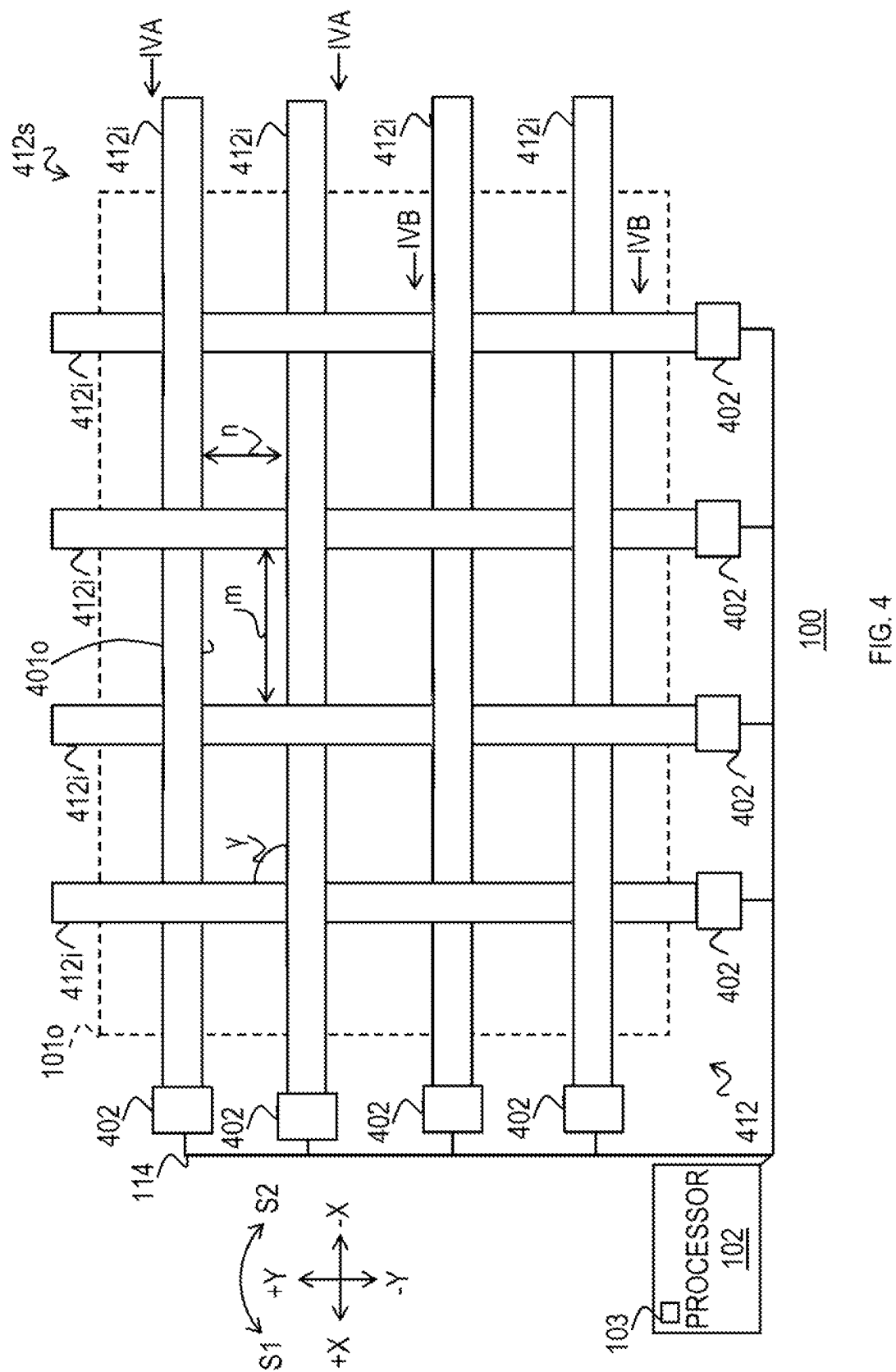


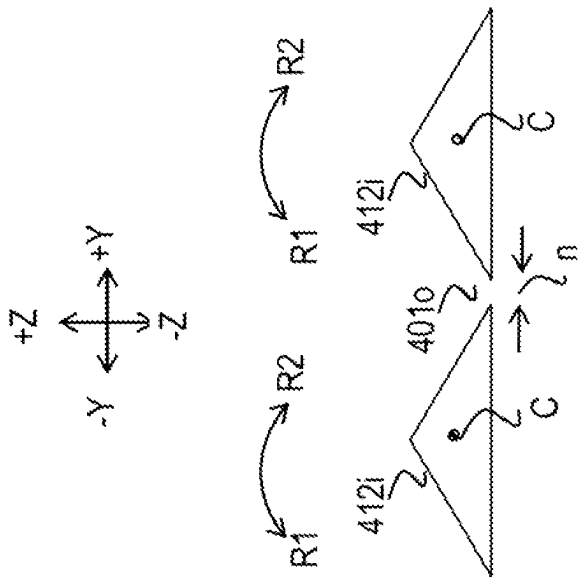
FIG. 2A



300

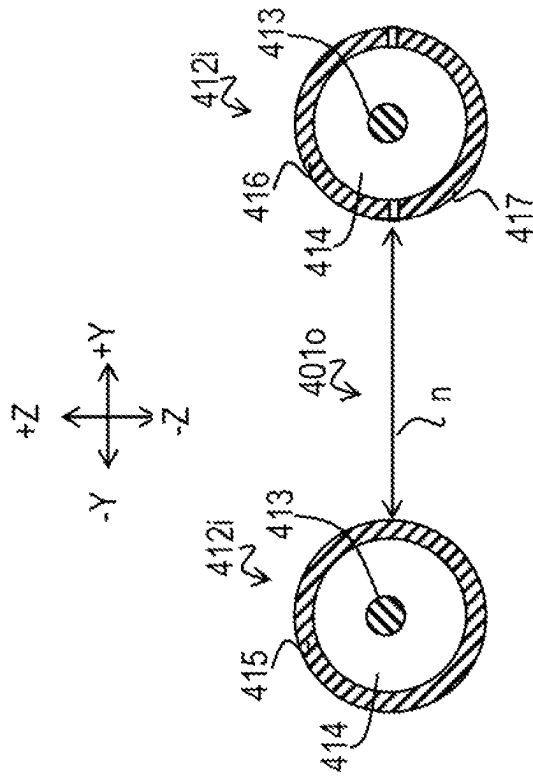
FIG. 3





412s

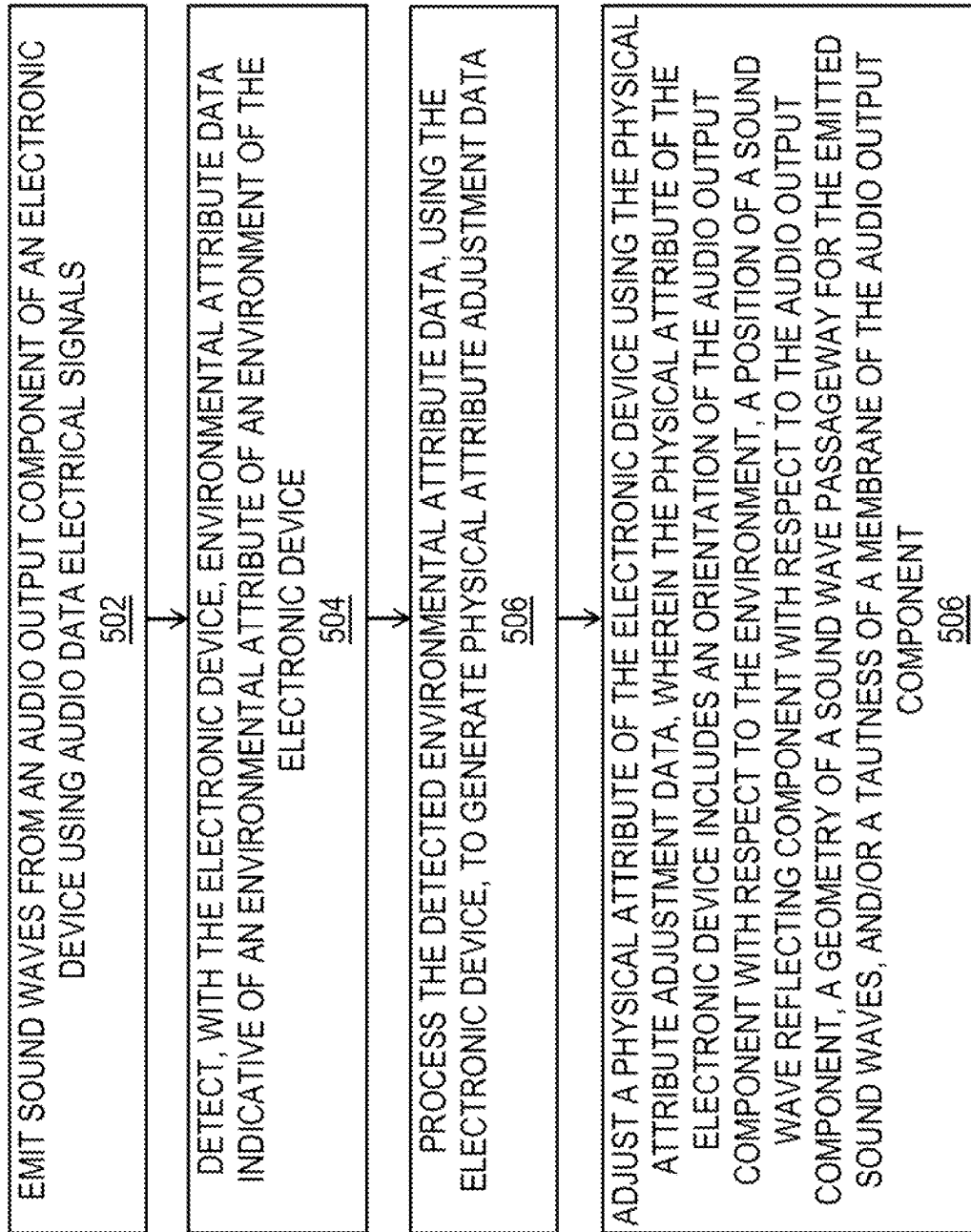
FIG. 4A



412s

FIG. 4B





500

FIG. 5

1

**ENHANCING A LISTENING EXPERIENCE  
BY ADJUSTING PHYSICAL ATTRIBUTES OF  
AN AUDIO PLAYBACK SYSTEM BASED ON  
DETECTED ENVIRONMENTAL  
ATTRIBUTES OF THE SYSTEM'S  
ENVIRONMENT**

CROSS-REFERENCE

This application is a continuation of patent application Ser. No. 16/932,549, filed Jul. 17, 2020, which is a continuation of patent application Ser. No. 16/355,128, filed Mar. 15, 2019, now U.S. Pat. No. 10,735,854, which is a continuation of patent application Ser. No. 15/657,844, filed Jul. 24, 2017, now U.S. Pat. No. 10,237,644, which claims the benefit of provisional patent application No. 62/398,900, filed Sep. 23, 2016, all of which are hereby incorporated by reference herein in their entireties.

FIELD

This generally relates to enhancing a listening experience and, more particularly, to enhancing a user's listening experience by adjusting physical attributes of an audio playback system based on detected environmental attributes of the system's environment.

BACKGROUND

Some user electronic devices may be operative to playback audio data for a listening user. However, the quality of the listening experience is often diminished by variables in the device's environment.

SUMMARY

Systems, methods, and computer-readable media are provided for enhancing a user's listening experience by adjusting physical attributes of an audio playback system based on detected environmental attributes of the system's environment.

As an example, a method of enhancing a listening experience of a user of an electronic device is provided that may include emitting sound waves from an audio output component of the electronic device using audio data electrical signals, detecting, with the electronic device, environmental attribute data indicative of an environmental attribute of an environment of the electronic device, processing the detected environmental attribute data, using the electronic device, to generate physical attribute adjustment data, and adjusting a physical attribute of the electronic device using the physical attribute adjustment data, wherein the physical attribute of the electronic device includes an orientation of the audio output component with respect to the environment, a position of a sound wave reflecting component with respect to the audio output component, a geometry of a sound wave passageway for the emitted sound waves, or a tautness of a membrane of the audio output component.

As an example, an electronic device is provided that may include a lower housing structure including an audio output component that emits sound waves into an environment of the electronic device, an upper housing structure including a display output component, a hinge structure coupling the lower housing structure to the upper housing structure, a sensor input component that detects environmental attribute data indicative of an environmental attribute of the environment of the electronic device, and a movement output

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component that adjusts the position of the upper housing structure with respect to the lower housing structure through rotation about the hinge structure based on the detected environmental attribute data for changing the reflection of the sound waves in the environment.

As yet another example, a product is provided that may include a non-transitory computer-readable medium and computer-readable instructions, stored on the computer-readable medium, that, when executed, are effective to cause a computer to detect environmental attribute data indicative of an environmental attribute of an ambient environment of the computer and adjust a physical attribute of the computer based on the environmental attribute data, wherein the physical attribute includes a position of an element of an audio output component of the computer with respect to the ambient environment of the computer, and wherein the environmental attribute includes geometry of the ambient environment, location of the user with respect to the audio output component, geometry of an ear of the user, and otoacoustic emission of an ear of the user.

This Summary is provided only to present some example embodiments, so as to provide a basic understanding of some aspects of the subject matter described in this document. Accordingly, it will be appreciated that the features described in this Summary are only examples and should not be construed to narrow the scope or spirit of the subject matter described herein in any way. Unless otherwise stated, features described in the context of one example may be combined or used with features described in the context of one or more other examples. Other features, aspects, and advantages of the subject matter described herein will become apparent from the following Detailed Description, Figures, and Claims.

BRIEF DESCRIPTION OF THE DRAWINGS

The discussion below makes reference to the following drawings, in which like reference characters refer to like parts throughout, and in which:

FIG. 1 is a schematic view of an illustrative audio playback system with an electronic device and at least one auxiliary assembly;

FIG. 2 is a perspective view of an exemplary electronic device and multiple auxiliary assemblies of the system of FIG. 1 in a particular system environment;

FIG. 2A is a cross-sectional view, taken from line IIA-IIA of FIG. 2, of a portion of the system of FIGS. 1 and 2; and

FIG. 3 is a schematic diagram of an example feedback loop of the system of FIGS. 1-2A;

FIG. 4 is a view of a portion of the device of the system of FIGS. 1, 2, and 2A;

FIG. 4A is a cross-sectional view, taken from line IVA-IVA of FIG. 4, of a portion of the device of FIGS. 1, 2, 2A, and 4;

FIG. 4B is a cross-sectional view, taken from line IVB-IVB of FIG. 4, of a portion of the device of FIGS. 1, 2, 2A, and 4; and

FIG. 5 is a flowchart of an illustrative process for enhancing a listening experience.

DETAILED DESCRIPTION

In the following detailed description, for purposes of explanation, numerous specific details are set forth to provide a thorough understanding of the various embodiments described herein. Those of ordinary skill in the art will realize that these various embodiments are illustrative only

and are not intended to be limiting in any way. Other embodiments will readily suggest themselves to such skilled persons having the benefit of this disclosure.

In addition, for clarity purposes, not all of the routine features of the embodiments described herein are shown or described. One of ordinary skill in the art will readily appreciate that in the development of any such actual embodiment, numerous embodiment-specific decisions may be required to achieve specific design objectives. These design objectives will vary from one embodiment to another and from one developer to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be a routine engineering undertaking for those of ordinary skill in the art having the benefit of this disclosure.

Systems, methods, and computer-readable media for enhancing a user's listening experience by adjusting physical attributes of an audio playback system based on detected environmental attributes of the system's environment are provided and described with reference to FIGS. 1-5.

FIG. 1 is a schematic view of an illustrative system 1 with an electronic device 100 and at least one auxiliary assembly 200, while FIGS. 2 and 2A are various views of a particular system 1 implemented within a particular environment E. Electronic device 100, on its own or in cooperation with one or more auxiliary assemblies 200, may be configured to detect various environmental attributes of the current environment of system 1 and to adjust various physical system attributes of system 1 based on the detected environmental attributes before or while a sound wave emitting subassembly of electronic device 100 emits sound waves into the environment of system 1, where such physical system attribute adjustment may enhance the experience of a system user listening to the emitted sound waves.

System 1 may be configured to detect any suitable environmental attributes of a current environment of system 1, including, but not limited to, the geometry (e.g., size and/or shape) of a room or defined space of the environment, the location and/or orientation of one or more system users within the environment relative to the sound wave emitting subassembly of device 100 (e.g., distance of a user from sound wave emitting subassembly and/or orientation of the ears with respect to the sound wave emitting subassembly), the specific identity or class identity of one or more system users within the environment, the geometry (e.g., size and/or shape) and/or the exposition of the ears of one or more system users within the environment relative to the sound wave emitting subassembly of device 100, the otoacoustic emissions (e.g., spontaneous otoacoustic emissions and/or evoked otoacoustic emissions) of the ears of one or more system users within the environment, the ambient noise level or other audio qualities of the environment distinct from any sound waves emitted by system 1, any audio qualities of the environment including the sound waves emitted by system 1, and/or the like. Electronic device 100 and/or any auxiliary assembly 200 of system 1 may include any suitable input component(s) (e.g., environmental attribute sensor input component(s)) that may be operative to detect any suitable environmental attribute of the environment of system 1 (e.g., cameras, ultrasonic sensors, infrared light sensors, microphones, temperature sensors, etc.) and/or may include any suitable communication component that may be operative to receive any suitable data indicative of any suitable environmental attribute of the environment of system 1 from any suitable remote data source (e.g., a data server (not shown) that may be operative to share data indicative of any suitable architectural characteristics of the

environment and/or data indicative of a particular user's ear structure or preferred audio equalization settings).

Before or while a sound wave emitting subassembly (e.g., any suitable transducer or driver that may be operative to receive audio data electrical signals and convert or transduce the received electrical signals into corresponding sound waves) of electronic device 100 may emit sound waves into the environment of system 1, system 1 may be configured to adjust, based on any detected environmental attributes of the environment of system 1, any suitable physical system attributes of system 1, including, but not limited to, the orientation of any element(s) of the sound wave emitting subassembly of device 100 with respect to any element(s) of the environment (e.g., the ears of a system user) in any one or more degrees of freedom (e.g., about any one or more axes of a three-dimensional Cartesian coordinate system for the environment), the geometry (e.g., size and/or shape) of any element(s) of the sound wave emitting subassembly of device 100, the location and/or orientation of any suitable sound wave reflecting component of device 100 and/or of any auxiliary assembly 200 relative to the sound wave emitting subassembly of device 100 and/or relative to any element(s) of the environment (e.g., the ears of a detected system user), the magnitude of any suitable movement (e.g., vibration, force, movement, actuator stroke, etc.) of any suitable movement output component, such as a movement output component embedded within or coupled to a sound wave reflecting component of device 100 and/or of any auxiliary assembly 200, and/or the like. In some embodiments, adjustment of one or more physical system attributes of system 1 may be based not only on any detected environmental attribute(s) of the environment of system 1 but also on any suitable characteristics of the sound waves emitted into the environment of system 1 by the sound wave emitting subassembly of device 100. Any physical system attribute adjustment may be made by system 1 to enhance the experience of a system user listening to the sound waves emitted by the sound wave emitting subassembly of device 100. Electronic device 100 and/or any auxiliary assembly 200 of system 1 may include any suitable output component(s) (e.g., physical or mechanical output components) that may be operative to be moved for adjusting any suitable physical system attributes of system 1 (e.g., sound reflecting surfaces, motors, piezoelectric actuators, etc.).

Electronic device 100 of system 1 may be any portable, wearable, mobile, or hand-held electronic device configured to emit sound waves, detect environmental attributes of its environment, and/or adjust physical attributes of system 1 to enhance a user's experience listening to the emitted sound waves. Alternatively, electronic device 100 may not be portable at all, but may instead be generally stationary. Electronic device 100 can include, but is not limited to, an audio player, game player, other media player, radio, medical equipment, domestic appliance, transportation vehicle instrument, musical instrument, cellular telephone (e.g., an iPhone™ available by Apple Inc.), other wireless communication device, personal digital assistant, remote control, pager, computer (e.g., a desktop, laptop, tablet, server, etc.), monitor, television, stereo equipment, set up box, set-top box, wearable device (e.g., an Apple Watch™ by Apple Inc.), boom box, modem, router, printer, and combinations thereof. Electronic device 100 may include any suitable control circuitry or processor 102, memory 104, communications component 106, power supply 108, input component 110, and output component 112. Electronic device 100 may also include a bus 114 that may provide one or more wired or wireless communication links or paths for transferring

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data and/or power to, from, or between various other components of device **100**. Device **100** may also be provided with a housing **101** that may at least partially enclose one or more of the components of device **100** for protection from debris and other degrading forces external to device **100**. In some embodiments, one or more of the components may be provided within its own housing (e.g., input component **110** may be an independent keyboard or mouse within its own housing that may wirelessly or through a wire communicate with processor **102**, which may be provided within its own housing). In some embodiments, one or more components of electronic device **100** may be combined or omitted. Moreover, electronic device **100** may include other components not combined or included in FIG. 1. For example, device **100** may include any other suitable components or several instances of the components shown in FIG. 1. For the sake of simplicity, only one of each of the components is shown in FIG. 1.

Memory **104** may include one or more storage mediums, including for example, a hard-drive, flash memory, permanent memory such as read-only memory (“ROM”), semi-permanent memory such as random access memory (“RAM”), any other suitable type of storage component, or any combination thereof. Memory **104** may include cache memory, which may be one or more different types of memory used for temporarily storing data for electronic device applications. Memory **104** may store media data (e.g., audio (e.g., music) and image and other media files), software (e.g., applications for implementing functions on device **100** (e.g., media playback applications and system environment processing applications)), firmware, preference information (e.g., media playback preferences), lifestyle information (e.g., food preferences), exercise information (e.g., information obtained by exercise monitoring equipment), transaction information (e.g., information such as credit card information), wireless connection information (e.g., information that may enable device **100** to establish a wireless connection), subscription information (e.g., information that keeps track of podcasts or television shows or other media a user subscribes to), contact information (e.g., telephone numbers and e-mail addresses), calendar information, any other suitable data, or any combination thereof.

Communications component **106** may be provided to allow device **100** to communicate with one or more other electronic devices or servers or subsystems (e.g., one or more auxiliary assemblies (e.g., assembly **200** of FIG. 1 and/or any one or more of assemblies **200a-200f** of FIGS. 2 and 2A)) using any suitable communications protocol(s). For example, communications component **106** may support Wi-Fi (e.g., an 802.11 protocol), Ethernet, Bluetooth™, near field communication (“NFC”), radio-frequency identification (“RFID”), high frequency systems (e.g., 900 MHz, 2.4 GHz, and 5.6 GHz communication systems), infrared, transmission control protocol/internet protocol (“TCP/IP”) (e.g., any of the protocols used in each of the TCP/IP layers), hypertext transfer protocol (“HTTP”), BitTorrent™, file transfer protocol (“FTP”), real-time transport protocol (“RTP”), real-time streaming protocol (“RTSP”), secure shell protocol (“SSH”), any other communications protocol, or any combination thereof. Communications component **106** may also include circuitry that can enable device **100** to be electrically coupled to another device or server or subsystem (e.g., one or more auxiliary assemblies **200**) and communicate with that other device, either wirelessly or via a wired connection (e.g., directly or via any suitable intermediate communication set-ups (e.g., servers, routers, towers, etc.)).

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Power supply **108** may provide power to one or more of the components of device **100**. In some embodiments, power supply **108** can be coupled to a power grid (e.g., when device **100** is not a portable device, such as a desktop computer). In some embodiments, power supply **108** can include one or more batteries for providing power (e.g., when device **100** is a portable device, such as a cellular telephone). As another example, power supply **108** can be configured to generate power from a natural source (e.g., solar power using solar cells).

One or more input components **110** may be provided to permit a user to interact or interface with device **100** (e.g., to provide any suitable user control data) and/or to detect any suitable environmental attributes of the environment of system **1** certain information about the ambient environment. For example, input component **110** can take a variety of forms, including, but not limited to, a touch pad, trackpad, dial, click wheel, scroll wheel, touch screen, one or more buttons (e.g., a keyboard), mouse, joy stick, track ball, switch, photocell, force-sensing resistor (“FSR”), encoder (e.g., rotary encoder and/or shaft encoder that may convert an angular position or motion of a shaft or axle to an analog or digital code), microphone, camera, scanner (e.g., a three-dimensional scanner that may identify the three-dimensional geometry (e.g., shape and/or size) of any suitable structure (e.g., the ear of a user), a barcode scanner or any other suitable scanner that may obtain product identifying information from a code, such as a linear barcode, a matrix barcode (e.g., a quick response (“QR”) code), or the like), proximity sensor (e.g., capacitive proximity sensor), biometric sensor (e.g., a fingerprint reader or other feature recognition sensor, which may operate in conjunction with a feature-processing application that may be accessible to electronic device **100** for authenticating or otherwise identifying or detecting a user), line-in connector for data and/or power, force sensor (e.g., any suitable capacitive sensors, pressure sensors, strain gauges, sensing plates (e.g., capacitive and/or strain sensing plates), etc.), ultrasonic sensor, thermal and/or temperature sensor (e.g., thermistor, thermocouple, thermometer, silicon bandgap temperature sensor, bimetal sensor, etc.) for detecting the temperature of a portion of electronic device **100** or an ambient environment thereof, a performance analyzer for detecting an application characteristic related to the current operation of one or more components of electronic device **100** (e.g., processor **102**), motion sensor (e.g., single axis or multi axis accelerometers, angular rate or inertial sensors (e.g., optical gyroscopes, vibrating gyroscopes, gas rate gyroscopes, or ring gyroscopes), linear velocity sensors, and/or the like), magnetometer (e.g., scalar or vector magnetometer), pressure sensor, light sensor (e.g., ambient light sensor (“ALS”), infrared (“IR”) sensor, etc.), acoustic sensor, sonic or sonar sensor, radar sensor, image sensor, video sensor, any suitable device locating subsystem or global positioning system (“GPS”) detector or subsystem, radio frequency (“RF”) detector, RF or acoustic Doppler detector, RF triangulation detector, electrical charge sensor, peripheral device detector, event counter, and any combinations thereof. Each input component **110** can be configured to provide one or more dedicated control functions for making selections or issuing commands associated with operating device **100**.

One or more output components **112** may be provided to present information (e.g., graphical, audible, and/or tactile information) to a user of device **100** and/or to adjust any physical system attribute of system **1**. For example, output component **112** can take a variety of forms, including, but not limited to, a sound wave emitting subassembly (e.g., any

suitable transducer or driver subassembly that may be operative to receive audio data electrical signals (e.g., of an audio or other suitable media file or streamed data that may be accessible to device **100**) and to convert or transduce the received electrical signals into corresponding sound waves), a sound wave reflecting subassembly (e.g., any suitable physical or mechanical sound wave reflecting component(s) that may be operative to reflect sound waves in any suitable manner) that may be moved in one or more directions (e.g., with respect to a sound wave emitting subassembly), any suitable physical or mechanical movement output component that may be operative to be moved for adjusting any suitable physical system attribute(s) of system **1** (e.g., motors, piezoelectric actuators, etc.) and that may be embedded within or coupled to a sound wave reflecting component or any other suitable component of device **100**, data and/or power line-out, visual display (e.g., for transmitting data via visible light and/or via invisible light), antenna, infrared port, flash (e.g., light sources for providing artificial light for illuminating an environment of the device), tactile/haptic component (e.g., rumblers, vibrators, etc.), taptic component (e.g., components that are operative to provide tactile sensations in the form of vibrations), and any combinations thereof.

It should be noted that one or more input components **110** and one or more output components **112** may sometimes be referred to collectively herein as an input/output (“I/O”) component or I/O interface **111** (e.g., input component **110** and display **112** as I/O component or I/O interface **111**). For example, input component **110** and display **112** may sometimes be a single I/O component **111**, such as a touch screen that may receive input information through a user’s touch of a display screen and that may also provide visual information to a user via that same display screen, or such as a transducer that may receive audio input information from a user when operating as a microphone and that may provide audio information to a user when operating as a speaker.

Processor **102** of device **100** may include any processing circuitry operative to control the operations and performance of one or more components of electronic device **100**. For example, processor **102** may be used to run one or more applications, such as an application **103**. Application **103** may include, but is not limited to, one or more operating system applications, firmware applications, media playback applications and/or environmental attribute processing applications and/or physical system attribute adjustment applications (e.g., a combined listening enhancement application), media editing applications, pass applications, calendar applications, state determination applications (e.g., device state determination applications, auxiliary assembly state determination applications), biometric feature-processing applications, compass applications, health applications, thermometer applications, weather applications, thermal management applications, force sensing applications, device diagnostic applications, video game applications, or any other suitable applications. For example, processor **102** may load application **103** as a user interface program or any other suitable program to determine how instructions or data received via an input component **110** and/or via any other component of device **100** (e.g., environmental attribute data or auxiliary assembly state/capability data from any auxiliary assembly **200** via communications component **106**, etc.) may manipulate the one or more ways in which information may be stored on device **100** (e.g., in memory **104**) and/or in which information may be provided to a user and/or in which physical system attributes may be adjusted via an output component **112** and/or in which auxiliary assembly

control data may be provided to a remote subsystem (e.g., to one or more auxiliary assemblies **200** via communications component **106**). Application **103** may be accessed by processor **102** from any suitable source, such as from memory **104** (e.g., via bus **114**) or from another device or server (e.g., from auxiliary assembly **200** via communications component **106** and/or from any other suitable remote data source (e.g., remote data server) via communications component **106**). Electronic device **100** (e.g., processor **102**, memory **104**, or any other components available to device **100**) may be configured to process data and/or generate commands at various resolutions, frequencies, and various other characteristics as may be appropriate for the capabilities and resources of device **100**. Processor **102** may include a single processor or multiple processors. For example, processor **102** may include at least one “general purpose” microprocessor, a combination of general and special purpose microprocessors, instruction set processors, audio processing units or sound cards, graphics processors, video processors, and/or related chips sets, and/or special purpose microprocessors. Processor **102** also may include on board memory for caching purposes. Processor **102** may be implemented as any electronic device capable of processing, receiving, or transmitting data or instructions. For example, processor **102** can be a microprocessor, a central processing unit, an application-specific integrated circuit, a field-programmable gate array, a digital signal processor, an analog circuit, a digital circuit, or combination of such devices. Processor **102** may be a single-thread or multi-thread processor. Processor **102** may be a single-core or multi-core processor. Accordingly, as described herein, the term “processor” may refer to a hardware-implemented data processing device or circuit physically structured to execute specific transformations of data including data operations represented as code and/or instructions included in a program that can be stored within and accessed from a memory. The term is meant to encompass a single processor or processing unit, multiple processors, multiple processing units, analog or digital circuits, or other suitably configured computing element or combination of elements.

Auxiliary assembly **200** may be any suitable assembly that may be configured to detect any suitable environmental attributes of the environment of system **1** and/or adjust any suitable physical system attributes of assembly **200**. Auxiliary assembly **200** may include any suitable control circuitry or processor **202**, which may be similar to any suitable processor **102** of device **100**, application **203**, which may be similar to any suitable application **103** of device **100**, memory **204**, which may be similar to any suitable memory **104** of device **100**, communications component **206**, which may be similar to any suitable communications component **106** of device **100**, power supply **208**, which may be similar to any suitable power supply **108** of device **100**, input component **210**, which may be similar to any suitable input component **110** of device **100**, output component **212**, which may be similar to any suitable output component **112** of device **100**, I/O interface **211**, which may be similar to any suitable I/O interface **111** of device **100**, bus **214**, which may be similar to any suitable bus **114** of device **100**, and/or housing **201**, which may be similar to any suitable housing **101** of device **100**. In some embodiments, one or more components of auxiliary assembly **200** may be combined or omitted. Moreover, auxiliary assembly **200** may include other components not combined or included in FIG. 1. For example, auxiliary assembly **200** may include any other suitable components or several instances of the components shown in FIG. 1. For the sake of simplicity, only one of each

of the components is shown in FIG. 1. Auxiliary assembly 200 may be operative to communicate any suitable data 91 (e.g., environmental attribute data detected by auxiliary assembly 200 (e.g., by any input component 210 of auxiliary assembly 200) and/or data indicative of the current state of any components/features of auxiliary assembly 200 and/or data indicative of any functionalities/capabilities of auxiliary assembly 200) from communications component 206 to communications component 106 of electronic device 100 using any suitable communication protocol(s), while electronic device 100 may be operative to communicate any suitable data 99 (e.g., auxiliary assembly control data operative to adjust any physical system attributes of auxiliary assembly 200 (e.g., of any output component(s) 212 of auxiliary assembly 200)) from communications component 106 to communications component 206 of auxiliary assembly 200 using any suitable communication protocol(s).

FIGS. 2 and 2A show system 1 implemented within a particular environment E, where system 1 may include electronic device 1 and various auxiliary assemblies 200a-200f, each of which may be similar to auxiliary assembly 200 and may include some or all of the components and/or functionality of assembly 200 of FIG. 1. As shown, environment E may include a space S at least partially defined by a back wall BW, a front wall (not shown), a left wall LW, a right wall RW, a floor FL, and a ceiling CL, where space may have a height H, a width W, and a depth P. Within space S, environment E may include a table T and other furniture N on floor FL, where electronic device 100 may be positioned on a top surface of table T. Moreover, as shown, environment E may include a first user U1 and a second user U2 within space S. It is to be appreciated that various elements of system 1 and/or environment E may not be to scale in FIG. 2 in order to clearly show certain features thereof. Assemblies 200a-200f may be positioned in any suitable manner throughout environment E, such as, for example, assembly 200a may be positioned about a side of electronic device 100, assembly 200b may be coupled to ceiling CL, assembly 200c may be coupled to left wall LW, assembly 200d may be worn by user U1, assembly 200e may be held by user U1, and assembly 200f may be resting on a top surface of furniture N but may be coupled to or configured as a drone or other suitable unmanned vehicle that may be moved or otherwise physically adjusted to any suitable position within space S.

As shown in FIGS. 2 and 2A, device 100 may be presented as a laptop or notebook personal computing device as an example only, while many other electronic devices (with or without displays) are envisioned. However, in FIGS. 2 and 2A, device 100 may include a “clamshell” form factor with a lower housing 101l, an upper housing 101u, and a hinge housing 101h that may rotatably couple lower housing 101l with upper housing 101u. Lower housing 101l may provide support for any suitable components, such as a left or first sound wave emitting subassembly output component 112a, a right or second sound wave emitting subassembly output component 112b, a sound wave reflecting output component 112c, a movement output component 112d (e.g., a piezoelectric actuator), and a keyboard input component 110a. Upper housing 101u may provide support for any suitable components, such as a camera input component 110b, a microphone input component 110c, a display output component 112e, and a movement output component 112f (e.g., a piezoelectric actuator). Camera input component 110b and/or any other suitable sensing input components of device 100 and/or of any auxiliary assembly of system 1 may be operative to detect any suitable

environmental attributes of environment E, such as the geometry (e.g., size and/or shape) of space S of environment E (e.g., height H, width W, and/or depth P), the location and/or orientation of user U1 and/or user U2 within environment E relative to sound wave emitting subassembly output component 112a and/or sound wave emitting subassembly output component 112b of device 100 (e.g., user U1 proximate and facing output component 112a and user U2 proximate yet facing away from output component 112b (e.g., in the direction M)), the specific identity or class identity of user U1 and/or user U2 within environment E, the geometry (e.g., size and/or shape) of the ears of user U1 and/or of user U2 (e.g., a three dimensional scan of the concha or other features of an ear that affect the frequency response of the ear) and/or the exposition of the ears of user U1 and/or of user U2 (e.g., the lack of exposition of the ears of user U2 due to user U2 wearing a winter hat H over the user's ears), and/or the like. Microphone input component 110c and/or any other suitable sensing input components of device 100 and/or of any auxiliary assembly of system 1 may be operative to detect any suitable environmental attributes of environment E, such as the otoacoustic emissions (e.g., spontaneous otoacoustic emissions and/or evoked otoacoustic emissions) of the ears of user U1 and/or user U2, the ambient noise level or other audio qualities of environment E distinct from any sound waves emitted by sound emitting subassembly output component 112a and/or by sound emitting subassembly output component 112b, any audio qualities of environment E including any sound waves emitted by system 1 (e.g., emitted sound wave SW and/or reflected sound wave SWR or any other sound waves within environment E), and/or the like.

Hinge housing 101h may provide support for any suitable components, such as a movement output component 112g, which may be operative to rotatably adjust (e.g., automatically without physical user interaction) the position of upper housing 101u with respect to lower housing 101l (e.g., to adjust the magnitude of angle  $\theta$  therebetween) such that one or more surfaces of at least a portion of upper housing 101h and/or display output component 112e or otherwise may also be operative to function as a sound wave reflecting subassembly for reflecting sound waves emitted from sound wave emitting subassembly output component 112a and/or from sound wave emitting subassembly output component 112b in any suitable direction (e.g., a magnitude of rotatable adjustment of the position of upper housing 101u with respect to the position of lower housing 101l and sound wave emitting subassembly output components 112a and 112b by movement output component 112g may be a physical system attribute that may be adjusted for enhancing a user's listening experience).

As shown in FIGS. 2 and 2A, sound wave emitting subassembly output component 112a may provide any suitable transducer or driver that may be operative to receive audio data electrical signals (e.g., from processor 102), to convert or transduce the received electrical signals into corresponding sound waves, and to emit the sound waves (e.g., sound waves SW) out from housing 101 through one or more audio housing openings 1010 and into environment E such that the sound waves (or reflections thereof (e.g., reflected sound waves SWR)) may be received at an eardrum of user U1 and/or user U2. As shown in FIG. 2A, sound wave emitting subassembly output component 112a may include a flexible diaphragm or membrane 152 that may be coupled at an outer periphery to a frame 154 and may include a former 152f at one or more intermediate positions with a moving coil 156 coupled thereto. A permanent

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magnet **158** may be positioned about moving coil **156**, for example, using frame **154**, at least one washer **157**, and a t-yoke **159**. The audio data electrical signals may be passed through coil **156** so as to generate an electromagnetic field that may produce an electromagnetic force that may be opposed by the main permanent magnetic field generated by permanent magnet **158** such that coil **156** may move membrane **152**, which may cause a disturbance in the air around membrane **152** for producing sound waves. At least some of these sound waves SW may be emitted through at least one audio housing opening **1010** of housing **101**. Therefore, membrane **152** may be operative to move in a magnetic gap for vibrating and producing sound waves. Membrane **152** may be any suitable shape and size, but may be a thin, semi-rigid but flexible structure. In some particular embodiments, membrane **152** may be a laminate or other suitable combination of multiple layers or films of materials stacked on top of one another to provide a composite structure that may be operative to provide or otherwise enable the tonality desired for sound wave emitting subassembly output component **112a** to generate a target sound.

As also shown in FIG. 2A, electronic device **100** may include a movement output component **112h** coupled to sound wave emitting subassembly output component **112a**, such as to a portion of frame **154**, where movement output component **112h** may be any suitable motor(s) or other suitable movement component(s) that may be operative to adjust any suitable physical attribute of sound wave emitting subassembly output component **112a** (e.g., a physical attribute other than that which may be adjusted by the audio data electrical signals passed through coil **156** for generating the sound waves to be emitted). For example, movement output component **112h** may receive any suitable physical system attribute adjustment data (e.g., from processor **102**) that may be operative to control movement output component **112h** to adjust the position and/or geometry of any suitable element(s) of sound wave emitting subassembly output component **112a**, such as moving the entirety of sound wave emitting subassembly output component **112a** up or down along an axis EA (e.g., to move sound wave emitting subassembly output component **112a** towards or away from housing opening **1010** of housing **101**), moving the entirety of sound wave emitting subassembly output component **112a** left or right along axis WA (e.g., to move sound wave emitting subassembly output component **112a** adjacent housing opening **1010** of housing **101**), rotating the entirety of sound wave emitting subassembly output component **112a** in either direction about axis EA (e.g., along path RP) or about axis WA or about another axis NA perpendicular to axes EA and WA, or the like, such that the entirety of sound wave emitting subassembly output component **112a** may be moved in any suitable manner with respect to housing opening **1010** of housing **101** for adjusting the orientation of any elements (e.g., membrane **152**) with respect to housing opening **1010** and ambient environment E (e.g., user U1). Alternatively or additionally, movement output component **112h** may receive any suitable physical system attribute adjustment data (e.g., from processor **102**) that may be operative to control movement output component **112h** to adjust the position and/or geometry of certain element(s) of sound wave emitting subassembly output component **112a** with respect to other elements of sound wave emitting subassembly output component **112a**, which may adjust an audio output characteristic of sound wave emitting subassembly output component **112a**, such as by moving outer periphery portion **152p1** of membrane **152** towards or away from outer periphery portion **152p2** of membrane **152** along

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axis MA for tightening or loosening membrane **152** (e.g., for adjusting the tautness of membrane **152** (e.g., the tautness of the sound wave generating element of output component **112a**)).

As also shown in FIG. 2A, electronic device **100** may include a movement output component/sound wave reflecting output component **112i** that may be operative to move a structure **112is** with respect to housing **101** for adjusting the shape and/or size and/or number of audio housing openings **1010** through which sound waves emitted by sound wave emitting subassembly output component **112a** may be able to travel. For example, structure **112is** may be moved in either direction along an axis OA for aligning each opening **1010** with a sound blocking portion of structure **112is** or with an audio structure opening **112io** through structure **112is**, where such alignment may either reduce the size of an audio housing opening **1010** through which sound waves may travel, taper or angle an orientation of an audio housing opening **1010** through which sound waves may travel (e.g., provide an angle to a passageway provided by a combination of an opening **112io** and an opening **1010**), or block an audio housing opening **1010**. Therefore, the geometry of structure **112is** and its openings **112io** and the position of structure **112is** (e.g., along axis OA) with respect to openings **1010** of housing **101** may be operative to adjust not only one or more physical system attributes of structure **112is** (e.g., its position within housing **101**) but also one or more physical system attributes of sound wave emitting subassembly output component **112a** (e.g., its geometry of sound wave passageways for emitting sound waves).

Additionally or alternatively, as shown in FIG. 4, electronic device **100** may include a movement output component/sound wave reflecting output component **412** that may be operative to adjust a geometry of a speaker grill structure **412s** of speaker grill elements **412i** that may be positioned above and/or under and/or within one or more audio housing openings **1010** for adjusting the shape and/or size and/or position of one or more structure openings **4010** between adjacent elements **412i** through which sound waves emitted by sound wave emitting subassembly output component **112a** may be able to travel for eventual receipt by one or more users. For example, structure **412s** may be a structure of any suitable number and arrangement of elements **412i** that may be operative to at least partially cover one or more audio housing openings **1010** for protecting sound wave emitting subassembly output component **112a** from debris or other potentially harmful forces in the environment of device **100**. As a particular example, as shown in FIG. 4, structure **412s** may include a four by four array of perpendicularly interlaced elements **412i** (e.g., an orthogonal mesh), although it is to be understood that any suitable number of elements **412i** may be provided in any suitable arrangement (e.g., crossing elements may not be interlaced over-under-over-under, as shown, but may be interlaced in any other suitable arrangement or may not be interlaced but may be laid on top of one another (e.g., all horizontal elements on top of all vertical elements, etc.). One, some, or each element **412i** may be made of any suitable material, such as metal, glass, rubber, polymer, fiber, and/or the like. One, some, or each element **412i** of structure **412s** may be coupled to an element adjustment component **402** of output component **412**, and each element adjustment component **402** may be controllable by processor **102** (e.g., via any suitable signals that may be communicated therebetween (e.g., via bus **114**)). An element adjustment component **402** may be controllable to adjust a shape, a size, and/or a position of an associated element **412i** of structure **412s**,

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which may adjust a shape, a size (e.g., dimension n), and/or a position of one or more structure openings **4010** that may be adjacent to and at least partially defined by the adjusted element **412i**.

Adjustment component(s) **402** may be controlled to move one or more elements **412i** with respect to one or more other elements **412i** within structure **412s** for adjusting any suitable physical characteristic of one or more openings **4010**. For example, an adjustment component **402** may receive any suitable physical system attribute adjustment data (e.g., from processor **102**) that may be operative to control that adjustment component **402** to adjust the position of its associated element **412i** in any suitable manner, such as by moving the entirety or at least a portion of element **412i** in the +X direction or the -X direction along an X-axis (e.g., to move a vertical element closer to or farther away from an adjacent vertical element (e.g., for adjusting a dimension m of one or more openings **4010**)), moving the entirety or at least a portion of element **412i** in the +Y direction or the -Y direction along a Y-axis (e.g., to move a horizontal element closer to or farther away from an adjacent horizontal element (e.g., for adjusting a dimension n of one or more openings **4010**)), moving the entirety or at least a portion of element **412i** in the +Z direction or the -Z direction along a Z-axis (e.g., to pull portions of an interlaced mesh closer to or farther away from output component **112a** and/or opening(s) **1010**), rotating the entirety or at least a portion of element **412i** in the S1 direction or the S2 direction about the Z-axis (e.g., to adjust the angular orientation of two or more elements (e.g., for adjusting the size of an angular dimension y between crossing elements)), rotating the entirety or at least a portion of element **412i** in the R1 direction or the R2 direction about the X-axis (e.g., to adjust the angular orientation of elements (e.g., rotating a horizontal element **412i** about its center C for adjusting the size of dimension n of opening **4010** between elements when a cross-sectional shape of one or more of the elements is non-circular (e.g., an isosceles triangle, as shown in FIG. 4A, or any other suitable shape that may adjust dimension n when rotated about center C))), adjusting the tension between ends of element **412i**, and/or the like, for adjusting any suitable physical characteristic of one or more openings **4010**, where adjustment component **402** may be any suitable motor(s) and/or any other suitable mechanisms that may physically move an associated element **412i** with respect to one or more other elements **412i** and/or opening(s) **1010** and/or output component **112**. Additionally or alternatively, an adjustment component **402** may receive any suitable physical system attribute adjustment data (e.g., from processor **102**) that may be operative to control that adjustment component **402** to adjust a cross-sectional geometry of its associated element **412i** in any suitable manner, such as by expanding or contracting a cross-sectional area of a horizontal element **412i** (e.g., in a Y-Z plane) by inflating or deflating a hollow portion of the element (e.g., with water or air or any other suitable fluid) and/or by adjusting an electrical field stimulating the element, and/or the like, for adjusting any suitable physical characteristic of one or more openings **4010** adjacent the element with the manipulated cross-section. As one particular example, as shown in FIG. 4B, an element **412i** may include an electrically conductive wire **413** extending along at least a portion of the length of the element that may be at least partially surrounded by an elastic material **414** (e.g., a low durometer silicone), which may be at least partially surrounded by an electrically conductive layer **415** (e.g., silver ink), such that when an electric field (e.g., differential charge) may be provided by component **402** via

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wire **413** and layer **415** to material **414**, material **414** may expand or contract, thereby changing the cross-sectional geometry of element **412i** (e.g., material **414** may be used as an electroactive polymer). In some embodiments, as also shown in FIG. 4B, two or more conductive layers **416** and **417** may be provided about different portions of material **414** of an element **412i**, such that different charges may be applied to different ones of layers **416** and **417** for adjusting the cross-sectional shape of element **412i** in various ways (e.g., such that the top half of the cross-sectional shape may not expand as much as the bottom half of the cross-sectional shape, such that the cross-sectional shape may be adjusted from a circular cross-sectional shape to a more triangular or other suitable shape, which may or may not be rotated as described with respect to FIG. 4A or otherwise moved with respect to one or more other elements **412i**), which may adjust the size and/or shape and/or taper angle of any opening **4010** of a sound wave passageway of device **100**. Therefore, the geometry of structure **412s** and its openings **4010** and the position of elements **412i** of structure **412s** with respect to opening(s) **1010** of housing **101** may be operative to adjust not only one or more physical system attributes of structure **412s** (e.g., the position of structure **412s** within housing **101** and/or the relative position and/or size and/or shape and/or orientation of different elements **412i** of structure **412s**) but also one or more physical system attributes of sound wave emitting subassembly output component **112a** (e.g., its geometry of sound wave passageways **4010** for emitting sound waves from device **100** into the environment).

As also shown in FIGS. 2 and 2A, electronic device **100** may include movement output component/sound wave reflecting output component **112c** that may be operative to move one or more structures **112cs** with respect to housing **101** for adjusting the location and/or orientation and/or position of one or more sound reflecting surfaces of structure(s) **112cs** relative to sound wave emitting subassembly output component **112a**, which may adjust the manner in which any sound waves emitted by sound wave emitting subassembly output component **112a** may be reflected by sound wave reflecting output component **112c** (e.g., adjust how sound wave SW may be reflected by a reflecting surface **112rs** of at least one structure **112cs** of output component **112c** as reflected sound wave SWR (e.g., adjust angle **1** of the reflection)). Various structures **112cs** and/or reflective surfaces of output component **112c** may be moved in any suitable manner (e.g., in any one or more degrees of freedom) with respect to output component **112a** (e.g., along a path LP about a hinge axis of component **112c** or in any direction along axis LA or axis FA or an axis NA perpendicular to axes LA and FA) for positioning one or more reflective surfaces in any suitable manner for any suitable reflection of sound waves (e.g., as determined by any suitable physical system attribute adjustment data received by component **112c** from processor **102**). It is to be appreciated that component **112c** may be configured to selectively be retracted into housing **101** (e.g., through housing opening **101c**) for hiding component **112c** when not in use.

As also shown, one or more reflective structures **112cs** of component **112c** may have embedded therein or otherwise coupled thereto one or more discrete movement output components **112cm** (e.g., a piezoelectric actuator), where each one of such movement output components **112cm** may be independently controlled (e.g., by any suitable physical system attribute adjustment data received processor **102**) to adjust the magnitude of a discrete movement of the movement component (e.g., a discrete vibration, etc.) that may be



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operative to affect any sound wave(s) reflecting off of the reflective structure **112c**s associated with the movement component. Similarly, movement component **112f** of device **100** (e.g., behind display output component **112e**) may be one or more discrete movement output components (e.g., a piezoelectric actuator), where each one of such movement output components may be independently controlled (e.g., by any suitable physical system attribute adjustment data received processor **102**) to adjust the magnitude of a discrete movement of the movement component (e.g., a discrete vibration, etc.) that may be operative to affect any sound wave(s) reflecting off of a reflective surface associated with the movement component (e.g., a surface of display output component **112e**). Similarly, movement component **112d** of device **100** (e.g., within housing structure **101i**) may be one or more discrete movement output components (e.g., a piezoelectric actuator) that may be independently controlled (e.g., by any suitable physical system attribute adjustment data received processor **102**) to adjust the magnitude of a discrete movement of the movement component (e.g., a discrete vibration, etc.) that may be operative to affect any sound wave(s) emitted by output component **112a** and/or to vibrate against table **T** for supplementing any sound wave(s) emitted by output component **112a**. Additionally, as shown, housing **101i** may include a microphone input component **110d** and/or any other suitable sensing input components that may be operative to detect any suitable environmental attributes of environment **E**, such as the otoacoustic emissions (e.g., spontaneous otoacoustic emissions and/or evoked otoacoustic emissions) of the ears of user **U1** and/or user **U2**, the ambient noise level or other audio qualities of environment **E** distinct from any sound waves emitted by sound emitting subassembly output component **112a** and/or by sound emitting subassembly output component **112b**, any audio qualities of environment **E** including any sound waves emitted by system **1** (e.g., emitted sound wave **SW** and/or reflected sound wave **SWR** or any other sound waves within environment **E**), and/or the like.

Auxiliary assembly **200a** may be removably coupled to a side of housing **101** of electronic device **100** and may include an output component **212a** that may be similar to movement output component/sound wave reflecting output component **112c**, with or without one or more discrete movement components, such that assembly **200a** may be operative to be positioned in any suitable manner to reflect or otherwise manipulate sound waves emitted from output component **112b** in any suitable manner. Similarly, auxiliary assembly **200b** may be coupled to ceiling **CL** and assembly **200c** may be coupled to left wall **LW** and assembly **200f** may be resting on a top surface of furniture **N**, each of which may be similar to movement output component/sound wave reflecting output component **112c**, with or without one or more discrete movement components, such that each assembly may be operative to be positioned in any suitable manner to reflect or otherwise manipulate any sound waves that may reach any suitable surface(s) of the assembly.

Auxiliary assembly **200d** may be worn by user **U1** in any suitable manner, such as about the user's head, such that different portions of assembly **200d** may physically interact with different portion of the user's head. For example, a first output component **212b** of assembly **200d** may be operative to be positioned adjacent user **U1**'s left ear such that physical system attribute adjustment of output component **212b** may physically manipulate the physical structure of user **U1**'s left ear (e.g., based on any suitable physical system attribute adjustment data **99** from device **100**, which may adjust the shape of the ear to better receive sound waves

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(e.g., to change the frequency response of the ear to enhance the listening experience of user **U1**)). Assembly **200d** may also include a microphone input component **210a** that may be operative to detect any suitable environmental attributes of environment **E**, such as the otoacoustic emissions (e.g., spontaneous otoacoustic emissions and/or evoked otoacoustic emissions) of the left ear of user **U1**, the ambient noise level or other audio qualities of environment **E** distinct from any sound waves emitted by sound emitting subassembly output component **112a** and/or by sound emitting subassembly output component **112b**, any audio qualities of environment **E** including any sound waves emitted by system **1** (e.g., emitted sound wave **SW** and/or reflected sound wave **SWR** or any other sound waves within environment **E**), and/or the like. Similarly a second output component **212c** of assembly **200d** may be operative to be positioned adjacent user **U1**'s right ear such that physical system attribute adjustment of output component **212c** may physically manipulate the physical structure of user **U1**'s right ear (e.g., based on any suitable physical system attribute adjustment data **99** from device **100**, which may adjust the shape of the ear to better receive sound waves (e.g., to change the frequency response of the ear to enhance the listening experience of user **U1**)). Assembly **200d** may also include a microphone input component **210b** that may be operative to detect any suitable environmental attributes of environment **E**, such as the otoacoustic emissions (e.g., spontaneous otoacoustic emissions and/or evoked otoacoustic emissions) of the right ear of user **U1**, the ambient noise level or other audio qualities of environment **E** distinct from any sound waves emitted by sound emitting subassembly output component **112a** and/or by sound emitting subassembly output component **112b**, any audio qualities of environment **E** including any sound waves emitted by system **1** (e.g., emitted sound wave **SW** and/or reflected sound wave **SWR** or any other sound waves within environment **E**), and/or the like. A third output component **212d** of assembly **200d** may be operative to be positioned against a back of user **U1**'s head as a discrete movement output component such that physical system attribute adjustment of output component **212d** may physically vibrate against the head of user **U1** in a particular manner to supplement the sensation of any sensed sound waves (e.g., based on any suitable physical system attribute adjustment data **99** from device **100**), which may enhance the listening experience of user **U1**). Assembly **200e** may be a handheld assembly of user **U1** (e.g., a smartphone) that may be operative to communicate any suitable data to device **100** (e.g., the identify of user **U1**, the location of user **U1**, the shape of each ear of user **U1** (e.g., if prompted to provided such information by device **100**), and/or the like.

Any one or more of assemblies **200a-200f** may include any other suitable output components that may be operative to adjust any suitable physical attribute of that assembly (e.g., based on any suitable physical system attribute adjustment data **99** from device **100**), such as a sound wave reflecting subassembly output component (e.g., any suitable physical or mechanical sound wave reflecting component(s) that may be operative to reflect sound waves in any suitable manner) and that may be moved in one or more directions within environment **E** (e.g., with respect to a sound wave emitting subassembly of device **100** and/or with respect to a user or otherwise), any suitable physical or mechanical movement output component that may be operative to be moved for adjusting any suitable physical system attribute(s) of the assembly (e.g., motors, piezoelectric actuators, etc.) and that may be embedded within or coupled to a sound

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wave reflecting component or any other suitable component of the assembly, and/or the like. Additionally or alternatively, each one of assemblies **200a-200f** may include any suitable input component that may be operative to detect any suitable environmental attribute(s) of environment E (e.g., for providing any suitable detected environmental attribute data **91** for use by device **100**).

Therefore, as may be illustrated in FIG. 3 by a schematic diagram **300** of an example feedback loop of system **1** of FIGS. 1-2A, processor **102** of device **100** (e.g., in conjunction with any other suitable processing of system **1** (e.g., by any processor **202** of any auxiliary assembly **200** or otherwise, which may be operative to also play back audio data therefrom)) may be operative to access audio data **93** representative of audio media to be played back by device **100** (e.g., from memory **104** or otherwise), any suitable desired (e.g., ideal) listening experience data **95** that may be indicative of preferred listening experience characteristics (e.g., for one or more particular users or for system **1** generally), such as sound wave frequency optimization, amplitude thresholds, and/or the like, and any suitable detected environment attribute data **91** (e.g., from any suitable input components **110** of device **100** and/or any suitable input components **210** of any auxiliary assembly **200** of system **1**, which may include one or more current physical system attributes of any suitable components of device **100** and/or of any assembly(ies) **200**) that may be indicative of the current environmental attributes of the environment of system **1**. Processor **102** may be operative to process such data **91**, **93**, and **95** (e.g., using any suitable application **103**) to generate appropriate physical system attribute adjustment data **99** that may be provided to any suitable output components **112** and/or output component **412** (e.g., to component(s) **402**) of device **100** and/or to any suitable output components **212** of any auxiliary assembly **200** of system **1** for adjusting one or more physical system attributes of system **1**. Processor **102** may also be operative to process such data **91**, **93**, and **95** (e.g., using any suitable application **103**) to generate appropriate audio data electrical signals **97** that may be applied to coils **156** of sound emitting subassembly output component **112a** and/or to coils of sound emitting subassembly output component **112b** for emitting sound waves indicative of audio data **93** that may then be received (e.g., without reflection or after reflection) by one or more users of the environment of system **1**. Then, new current environmental attributes of the environment of system **1** may be detected by input components **110/210** and provided as data **91** to processor **102** for processing in order to potentially update signals **97** and **99**. Therefore, system **1** may be operative to detect various environmental attributes of the current environment of system **1** and to adjust various physical system attributes of system **1** based on the detected environmental attributes before or while a sound wave emitting subassembly of electronic device **100** emits sound waves into the environment of system **1**, where such physical system attribute adjustment may enhance the experience of a system user listening to the emitted sound waves (e.g., by comparing actual environmental attributes with desired listening attributes of data **95** to reduce the error therebetween for achieving and maintaining a desired output condition). In the case of multiple users, as shown in FIG. 2, adjustments may be made to enhance the experience of each user (e.g., an adjustment of component **112b** may be made to enhance the experience of user U2 while adjustment of component **112a** may be made to enhance the experience of user U1).

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FIG. 5 is a flowchart of an illustrative process **500** for enhancing a listening experience of a user of an electronic device. At operation **502** of process **500**, sound waves may be emitted waves from an audio output component of the electronic device using audio data electrical signals. At operation **504** of process **500**, the electronic device may detect environmental attribute data indicative of an environmental attribute of an environment of the electronic device. At operation **506** of process **500**, a physical attribute of the electronic device may be adjusted using the physical attribute adjustment data, wherein the physical attribute of the electronic device includes at least one of the following: an orientation of the audio output component with respect to the environment; a position of a sound wave reflecting component with respect to the audio output component; a geometry of a sound wave passageway for the emitted sound waves; and a tautness of a membrane of the audio output component.

It is understood that the operations shown in process **500** of FIG. 5 are only illustrative and that existing operations may be modified or omitted, additional operations may be added, and/or the order of certain operations may be altered.

Moreover, the processes described with respect to FIGS. 1-5, as well as any other aspects of the disclosure, may each be implemented by software, but may also be implemented in hardware, firmware, or any combination of software, hardware, and firmware. They each may also be embodied as computer-readable code recorded on a computer-readable medium. The computer-readable medium may be any data storage device that can store data or instructions which can thereafter be read by a computer system. Examples of the computer-readable medium may include, but are not limited to, read-only memory, random-access memory, flash memory, CD-ROMs, DVDs, magnetic tape, and optical data storage devices (e.g., memory **104** and/or memory **204** of FIG. 1). The computer-readable medium can also be distributed over network-coupled computer systems so that the computer readable code is stored and executed in a distributed fashion. For example, the computer-readable medium may be communicated from one electronic device to another electronic device using any suitable communications protocol (e.g., the computer-readable medium may be communicated to electronic device **100** via communications component **106**). The computer-readable medium may embody computer-readable code, instructions, data structures, program modules, or other data in a modulated data signal, such as a carrier wave or other transport mechanism, and may include any information delivery media. A modulated data signal may be a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal.

It is to be understood that program modules and/or various processes or operations of system **1** may be provided as a software construct, firmware construct, one or more hardware components, or a combination thereof. For example, various processes or operations or modules of system **1** may be described in the general context of computer-executable instructions, such as program modules, that may be executed by one or more computers or other devices. Generally, a program module may include one or more routines, programs, objects, components, and/or data structures that may perform one or more particular tasks or that may implement one or more particular abstract data types. It is also to be understood that the number, configuration, functionality, and interconnection of the modules are merely illustrative, and that the number, configuration, functionality, and interconnection of existing modules may be modi-

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fied or omitted, additional modules may be added, and the interconnection of certain modules may be altered.

At least a portion of one or more of the processes or operations or modules of system **201** may be stored in or otherwise accessible to device **100** in any suitable manner (e.g., in memory **104** of device **100** or via communications component **106** of device **100** and/or in memory **204** of device **200** or via communications component **206** of device **200**). Each module of system **201** may be implemented using any suitable technologies (e.g., as one or more integrated circuit devices), and different modules may or may not be identical in structure, capabilities, and operation. Any or all of the processes or operations or modules or other components of system **201** may be mounted on an expansion card, mounted directly on a system motherboard, or integrated into a system chipset component (e.g., into a “north bridge” chip). System **201** may include any amount of dedicated sound processing memory.

Many alterations and modifications of the preferred embodiments will no doubt become apparent to a person of ordinary skill in the art after having read the foregoing description, it is to be understood that the particular embodiments shown and described by way of illustration are in no way intended to be considered limiting. Thus, references to the details of the described embodiments are not intended to limit their scope. Therefore, obvious substitutions now or later known to one with ordinary skill in the art are defined to be within the scope of the defined elements. It is also to be understood that various directional and orientational terms, such as “up” and “down,” “front” and “back,” “exterior” and “interior,” “top” and “bottom” and “side,” “length” and “width” and “depth,” “thickness” and “diameter” and “cross-section” and “longitudinal,” “X-” and “Y-” and “Z-,” and the like may be used herein only for convenience, and that no fixed or absolute directional or orientational limitations are intended by the use of these words.

What is claimed is:

1. An electronic device, comprising:  
wireless communications circuitry that is configured to wirelessly receive user location information;  
a speaker; and  
a motor that is configured to adjust the speaker based on the user location information, wherein the motor is configured to adjust a diaphragm of the speaker.
2. The electronic device defined in claim 1, wherein the motor is configured to adjust a tautness of the diaphragm of the speaker.
3. The electronic device defined in claim 1, wherein the motor is configured to move a portion of the diaphragm relative to a housing of the electronic device.
4. The electronic device defined in claim 1, wherein the motor is configured to move a sound-blocking structure that overlaps the speaker.
5. The electronic device defined in claim 1, wherein the user location information is transmitted by an auxiliary electronic device and wherein the electronic device receives information indicating a distance between the electronic device and the auxiliary electronic device.
6. The electronic device defined in claim 1, wherein the user location information is transmitted by an auxiliary

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electronic and wherein the electronic device receives information indicating a current state of the auxiliary electronic device.

7. The electronic device defined in claim 6, wherein the auxiliary electronic device comprises a cellular telephone.

8. The electronic device defined in claim 6 wherein the electronic device receives a geometry of an environment from the auxiliary electronic device.

9. The electronic device defined in claim 6 wherein the electronic device receives user identity information from the auxiliary electronic device.

10. The electronic device defined in claim 6 wherein the electronic device receives ear geometry information from the auxiliary electronic device.

11. An electronic device, comprising:  
wireless communications circuitry that wirelessly receives transmitted user location information;  
a display;  
a speaker that emits sound waves in a first direction; and  
a motor, wherein the motor is configured to adjust a position of the display relative to the speaker based on the user location information.

12. The electronic device defined in claim 11, wherein the speaker is mounted in a housing of the electronic device and wherein the motor is configured to move a structure of the electronic device that is mounted outside the housing.

13. The electronic device defined in claim 11, wherein the wireless communications circuitry wirelessly receives distance information indicating a distance between the electronic device and an auxiliary electronic device.

14. The electronic device defined in claim 13, wherein the user location information is transmitted by the auxiliary electronic device.

15. An electronic device, comprising:  
wireless communications circuitry that wirelessly receives transmitted user location information;  
a speaker that emits sound waves in a first direction; and  
a motor, wherein the motor is configured to adjust the speaker to emit the sound waves in a second direction based on the user location information and wherein the motor is configured to adjust a tautness of a membrane of the speaker.

16. An electronic device, comprising:  
a housing;  
a speaker that emits sound waves, wherein the speaker is mounted in the housing;  
wireless communications circuitry that receives information indicating a user location; and  
a discrete movement structure that moves based on the user location, wherein the discrete movement structure is coupled to a display and is configured to move the display relative to the speaker.

17. The electronic device defined in claim 16, wherein the discrete movement structure moves by rotating about a hinge structure.

18. The electronic device defined in claim 16, wherein the discrete movement structure moves along a first axis.

19. The electronic device defined in claim 18, wherein the discrete movement structure moves along a second axis that extends at a non-zero angle to the first axis.

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